Week 10: Gibbs Sampling for Mixture Models

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1 Clustering and Mixture Models

Suppose we observe data $y_1, \dots, y_n \in \mathbb{R}^d$ and believe that they arise from several distinct subpopulations. The goal of *clustering* is to infer these groups from the observations. Unlike classification, no class labels are observed; clustering is therefore an example of *unsupervised learning*.

1.1 From K-means to a probabilistic model

The K-means algorithm uses centres $\mu_1, \dots, \mu_K$ and labels $z_i \in \{1, \dots, K\}$, where $z_i = j$ means that observation i is assigned to group j. It seeks to minimize the within-cluster sum of squares

$$\sum_{i=1}^{n} \|y_i - \mu_{z_i}\|^2$$

by alternating between assigning each observation to its nearest centre and replacing each centre by the average of its assigned observations. These updates do not increase the objective, but need not find its global minimum.

A probabilistic model gives us a way to describe uncertainty about both the groups and their parameters. To keep the calculations simple, we consider real-valued observations from now on. Suppose that observation i comes from population j with probability w_j, and that population j has distribution $N(\mu_j, \sigma_j^2)$. We can express this as

$$z_i \mid w \sim \text{Categorical}(w_1, \dots, w_K), \qquad y_i \mid z_i = j, \mu, \sigma^2 \sim N(\mu_j, \sigma_j^2),$$

independently across observations conditional on the parameters. Here $w_j \geq 0$ and $\sum_j w_j = 1$. Summing over the unobserved label gives

$$f(y_i \mid w, \mu, \sigma^2) = \sum_{j=1}^{K} w_j \phi(y_i; \mu_j, \sigma_j^2),$$

where $\phi(y; \mu, \sigma^2)$ denotes the normal density. This is a *Gaussian mixture model*. The labels have the same interpretation as in K-means, but now we can ask how probable each assignment is, rather than making only a single assignment.

2 Gibbs Sampler for Mixture Models

2.1 Why is posterior inference difficult?

Start with two populations having known unit variances and a known mixing proportion $p \in (0,1)$:

$$y_i \overset{\text{i.i.d.}}{\sim} (1-p)N(\mu_0, 1) + pN(\mu_1, 1), \qquad i = 1, \ldots, n. \tag{1}$$

Place independent $N(0, C)$ priors on μ_0 and μ_1, with a fixed, finite $C > 0$. Writing $y = (y_1, \ldots, y_n)$, the posterior is

$$\pi(\mu_0, \mu_1 \mid y) \propto \exp\left\{-\frac{\mu_0^2 + \mu_1^2}{2C}\right\} \prod_{i=1}^{n} \left\{(1-p)\phi(y_i; \mu_0, 1) + p\phi(y_i; \mu_1, 1)\right\}.$$

The obstacle is the *product of sums*. For a single normal population, multiplying the likelihood by a normal prior leaves a quadratic expression in the mean, so the posterior is normal. Here each observation could belong to either population, and its likelihood contribution couples the two means. Even if we fix one mean, the conditional posterior of the other is not a single normal distribution. Thus the usual normal–normal calculation cannot be applied directly.

We can evaluate the displayed density up to its normalizing constant, but direct posterior sampling and integration are not straightforward. Expanding the product would produce 2^n terms, one for each possible assignment of observations to the two populations. Although numerical grid integration is possible in this two-parameter example, it becomes impractical as the number of components and unknown parameters grows.

The difficulty suggests its own solution: *if we knew which population generated each observation, we would have two ordinary normal inference problems.* Gibbs sampling exploits exactly this observation, while allowing the unknown assignments to vary.

2.2 Introducing labels makes the updates simple

Introduce independent latent indicators $z_i \sim \text{Bernoulli}(p)$, with $y_i \mid z_i = j, \mu_j \sim N(\mu_j, 1)$ for $j = 0, 1$. This gives precisely the marginal model (1). We now sample from the enlarged posterior $\pi(\mu_0, \mu_1, z \mid y)$, where $z = (z_1, \ldots, z_n)$. Keeping just the mean coordinates of these draws then gives draws from the posterior of interest.

The joint posterior of the two means and all the latent labels follows from Bayes' theorem. The prior density of the means is $\phi(\mu_0; 0, C)\phi(\mu_1; 0, C)$, the prior mass of the labels is $\prod_{i=1}^{n} p^{z_i}(1-p)^{1-z_i}$, and the conditional likelihood is $\prod_{i=1}^{n} \phi(y_i; \mu_{z_i}, 1)$. Consequently,

$$\pi(\mu_0, \mu_1, z \mid y) = \frac{1}{m(y)}\, \phi(\mu_0; 0, C)\phi(\mu_1; 0, C) \prod_{i=1}^{n} \left[p^{z_i}(1-p)^{1-z_i}\phi(y_i; \mu_{z_i}, 1)\right], \qquad z \in \{0,1\}^n. \tag{2}$$

where $m(y)$ is the marginal density of the observations (the normalizing constant). Unlike the original mixture likelihood, the augmented likelihood contains no sums over components. Conditional on the labels, it factors into two ordinary Gaussian likelihoods.

Summing this joint posterior over all label vectors z recovers the marginal posterior $\pi(\mu_0, \mu_1 \mid y)$ displayed above: the sum factorizes into the product of the two-component

mixture likelihoods. Thus introducing labels does not change the posterior inference about the means; it makes the augmented posterior easier to sample from.

We now obtain the two full conditionals directly from (2).

1. Sample the labels given the means. By Bayes' rule,

$$r_i := P(z_i = 1 \mid \mu_0, \mu_1, y) = \frac{p\phi(y_i; \mu_1, 1)}{(1-p)\phi(y_i; \mu_0, 1) + p\phi(y_i; \mu_1, 1)}. \tag{3}$$

Given the means and data, the labels are independent: the joint likelihood and label prior factor across observations. We therefore sample each z_i independently from Bernoulli(r_i). An observation is more likely to be assigned to a component when that component gives it a larger weighted density, but uncertain assignments remain random.

2. Sample the means given the labels. Let $n_j = \sum_i I\{z_i = j\}$ and $S_j = \sum_{i:z_i=j} y_i$ for $j = 0, 1$. Conditional on the labels, component j receives exactly the observations assigned to it. Its conditional posterior kernel is

$$\pi(\mu_j \mid z, y) \propto \exp\left\{-\frac{\mu_j^2}{2C} - \frac{1}{2}\sum_{i:z_i=j}(y_i - \mu_j)^2\right\}.$$

This is the familiar normal–normal conjugate calculation. The posterior precision is the sum of prior precision C^{-1} and data precision n_j, namely $n_j + C^{-1}$. Completing the square gives, independently for $j = 0, 1$,

$$\mu_j \mid z, y \sim N\left(\frac{S_j}{n_j + C^{-1}}, \frac{1}{n_j + C^{-1}}\right). \tag{4}$$

When $n_j > 0$ and C is large, the conditional mean is close to the sample average in group j, and the conditional variance is close to $1/n_j$. Thus this update is just Bayesian inference for a normal mean using the currently assigned observations. If $n_j = 0$, then $S_j = 0$ and the conditional posterior is simply the prior $N(0, C)$.

The Gibbs sampler. Initialize $\mu_0^{(0)}, \mu_1^{(0)}$. At iteration t:

1. Evaluate (3) at the current means and sample all labels $z_i^{(t+1)}$ independently.
2. Form n_j and S_j from these new labels, then sample $\mu_0^{(t+1)}, \mu_1^{(t+1)}$ independently using (4).

Each step samples a full conditional of the enlarged posterior, so a complete sweep preserves that posterior by the Gibbs sampling principle. Under the usual convergence conditions, averages along the chain approximate posterior expectations.

Remark 2.1 (The main takeaway)**.** *A difficult marginal posterior can become computationally simple after introducing latent variables. In the Gaussian mixture model, the posterior of* (μ_0, μ_1) *is complicated because the likelihood is a product of sums. Introducing the unobserved population labels* $z_1, \ldots, z_n$ *makes the full conditional distributions simple:*

- *Given the means, the labels are independent Bernoulli random variables, with probabilities obtained by Bayes' rule.*

- *Given the labels, the means have independent normal posteriors, obtained by normal–normal conjugacy.*

Gibbs sampling alternates between these two familiar calculations. Thus, somewhat surprisingly, adding n unknown variables makes posterior computation easier.

Unlike K-means, which produces a single assignment of observations to clusters, Gibbs sampling continually resamples both the labels and the means. The resulting posterior inference therefore incorporates uncertainty about cluster assignments as well as the unknown population means.

Questions for discussion.

1. Why is direct posterior sampling difficult in the Gaussian mixture model? What is the mathematical obstacle in the marginal likelihood?
2. Why does introducing n additional unknown labels make posterior computation easier rather than harder? Which familiar conjugate calculations become available?
3. How is Gibbs sampling for a Gaussian mixture model different from K-means clustering? In particular, how does Gibbs sampling account for uncertainty about cluster assignments and population means?

Remark 2.2 (Initialization, local modes, and slow mixing)**.** *Although Gibbs sampling preserves the correct posterior, it need not explore that posterior efficiently. Mixture-model posteriors can have multiple modes corresponding to different cluster configurations. A chain initialized near one mode may remain there for a long time because moving elsewhere may require coordinated changes in labels and means.*

Consequently, finite-run posterior summaries may depend substantially on initialization. Running chains from different starting points and comparing their trajectories and summaries is a useful diagnostic. The distinction is between having the correct stationary distribution and reaching it efficiently: under suitable ergodicity conditions, initialization does not change the limiting posterior averages, but it can strongly affect the accuracy of a finite simulation.

2.3 Unknown weights and variances: the same conjugate calculations

The same approach works when the mixing weights and component variances are unknown. For a fixed number K of components, write

$$y_i \overset{\text{i.i.d.}}{\sim} \sum_{j=1}^{K} w_j N(\mu_j, \sigma_j^2), \qquad w = (w_1, \ldots, w_K), \quad \mu = (\mu_1, \ldots, \mu_K), \quad \sigma^2 = (\sigma_1^2, \ldots, \sigma_K^2).$$

Use independent priors

$$w \sim \text{Dirichlet}(a_1, \ldots, a_K), \qquad \mu_j \sim N(m, s^2), \qquad \sigma_j^2 \sim \text{IG}(\alpha, \beta), \tag{5}$$

where $a_j, \alpha, \beta, s^2 > 0$. Our inverse-gamma convention is $p(x) \propto x^{-\alpha-1} \exp(-\beta/x)$ for $x > 0$. For two components, the Dirichlet prior is equivalent to a beta prior on one of the mixing proportions. The independent normal and inverse-gamma priors give standard *full conditional* distributions; we update the means and variances in separate steps.

Retain the latent representation introduced above, now with $z_i \in \{1, \dots, K\}$. Once the labels are given, the *complete-data likelihood* groups the observations by component:

$$p(y, z \mid w, \mu, \sigma^2) = \prod_{j=1}^{K} \left[w_j^{n_j} \prod_{i : z_i = j} \phi(y_i; \mu_j, \sigma_j^2) \right], \qquad n_j = \sum_{i=1}^{n} I\{z_i = j\}.$$

Multiplying by the priors gives the joint posterior kernel

$$\begin{aligned} \pi(w, \mu, \sigma^2, z \mid y) \propto \prod_{j=1}^{K} \Bigg[w_j^{a_j + n_j - 1} \exp\left\{ -\frac{(\mu_j - m)^2}{2s^2} \right\} (\sigma_j^2)^{-\alpha - 1 - n_j/2} \\ \times \exp\left\{ -\frac{\beta + \frac{1}{2} \sum_{i : z_i = j} (y_i - \mu_j)^2}{\sigma_j^2} \right\} \Bigg]. \end{aligned} \tag{6}$$

The weights lie on the probability simplex and the variances are positive. To find a full conditional, we simply retain the factors involving the variable being updated and recognize a familiar density.

Labels: Bayes' rule. As in the two-component example, the labels are conditionally independent, with probabilities

$$r_{ij} := P(z_i = j \mid y, w, \mu, \sigma^2) = \frac{w_j \phi(y_i; \mu_j, \sigma_j^2)}{\sum_{\ell=1}^{K} w_\ell \phi(y_i; \mu_\ell, \sigma_\ell^2)}. \tag{7}$$

These probabilities are called *responsibilities*: r_{ij} measures the conditional probability that component j generated observation i. Sample z_i from the categorical distribution with probabilities $(r_{i1}, \dots, r_{iK})$.

Weights: Dirichlet–multinomial conjugacy. Given the labels, we have n_j observations in category j. The terms involving w in (6) are $\prod_j w_j^{a_j + n_j - 1}$, so

$$w \mid z, y, \mu, \sigma^2 \sim \text{Dirichlet}(a_1 + n_1, \dots, a_K + n_K). \tag{8}$$

This is exactly the usual update of Dirichlet prior parameters by observed category counts.

Means: normal–normal conjugacy. Given the labels and variances, component j is a normal sample with known variance σ_j^2. The terms involving μ_j are

$$\pi(\mu_j \mid z, y, \sigma_j^2) \propto \exp\left\{ -\frac{(\mu_j - m)^2}{2s^2} - \frac{1}{2\sigma_j^2} \sum_{i : z_i = j} (y_i - \mu_j)^2 \right\}.$$

Thus the means are conditionally independent, with

$$\begin{aligned} \mu_j \mid z, y, \sigma_j^2 &\sim N(M_j, V_j), \\ V_j = \left(\frac{1}{s^2} + \frac{n_j}{\sigma_j^2} \right)^{-1}, &\qquad M_j = V_j \left(\frac{m}{s^2} + \frac{S_j}{\sigma_j^2} \right). \end{aligned} \tag{9}$$

where $S_j = \sum_{i : z_i = j} y_i$. Prior precision and sample precision add, and the posterior mean balances the prior mean with the assigned data.

Variances: inverse-gamma conjugacy. Given the labels and means, the terms involving σ_j^2 are

$$\pi(\sigma_j^2 \mid z, y, \mu) \propto (\sigma_j^2)^{-(\alpha+n_j/2)-1} \exp\left\{-\frac{\beta + \frac{1}{2}\sum_{i:z_i=j}(y_i-\mu_j)^2}{\sigma_j^2}\right\}.$$

Recognizing the inverse-gamma kernel gives, independently across j,

$$\sigma_j^2 \mid z, y, \mu \sim \text{IG}\left(\alpha + \frac{n_j}{2}, \beta + \frac{1}{2}\sum_{i:z_i=j}(y_i-\mu_j)^2\right). \tag{10}$$

Again this is the familiar normal variance calculation: half the sample size updates the shape parameter, and half the residual sum of squares updates the scale parameter.

One complete sweep. Initialize weights $w_j^{(0)} > 0$ summing to one, means $\mu_j^{(0)}$, and variances $(\sigma_j^2)^{(0)} > 0$. At iteration t:

1. Use the current parameters in (7) to sample $z^{(t+1)}$, then compute the counts n_j and sums S_j.
2. Sample $w^{(t+1)}$ using (8) and the new counts.
3. Sample $\mu_j^{(t+1)}$ using (9), with the new labels and the current variances $(\sigma_j^2)^{(t)}$.
4. Sample $(\sigma_j^2)^{(t+1)}$ using (10), with the new labels and the newly sampled means $\mu_j^{(t+1)}$.

The full sampler requires only categorical, Dirichlet, normal and inverse-gamma draws. The main computational obstacle was the missing labels; conditional on them, all the parameter updates come from conjugate calculations we already know.

2.4 What do the posterior draws tell us?

After allowing the chain to explore the posterior, retain T draws, indexed below by $t = 1, \ldots, T$. These draws describe uncertainty about the mixture and its clustering, and can be used for more than a single parameter estimate.

For example, the posterior predictive density for a new observation is

$$p(y_{\text{new}} \mid y) = E\left[\sum_{j=1}^{K} w_j \phi(y_{\text{new}}; \mu_j, \sigma_j^2) \,\middle|\, y\right] \approx \frac{1}{T}\sum_{t=1}^{T}\sum_{j=1}^{K} w_j^{(t)} \phi(y_{\text{new}}; \mu_j^{(t)}, (\sigma_j^2)^{(t)}). \tag{11}$$

Each draw supplies a possible mixture density; averaging them accounts for posterior uncertainty in its weights, means and variances.

For clustering, we can estimate the posterior probability that observation i belongs to population ℓ:

$$P(z_i = \ell \mid y) \approx \frac{1}{T}\sum_{t=1}^{T} I\{z_i^{(t)} = \ell\}.$$

We can assign each observation to the population with the largest posterior probability. These probabilities also quantify uncertainty: a probability close to one indicates a confident assignment, whereas similar probabilities across several populations indicate ambiguity.

Remark 2.3 (Component labels and clustering summaries)**.** *When the priors treat all components symmetrically, permuting the component labels does not change the model or posterior density. A sampler may therefore exchange the numerical labels of the same clusters. Component-specific probabilities and parameter summaries require a consistent labeling convention or an appropriate relabeling of the draws. A summary that avoids this issue is the probability that two observations belong to the same group:*

$$P(z_i = z_h \mid y) \approx \frac{1}{T} \sum_{t=1}^{T} I\{z_i^{(t)} = z_h^{(t)}\}.$$

This probability is unchanged by any permutation of component labels.

The EM algorithm uses the same latent-variable idea for maximum likelihood estimation. Gibbs sampling draws labels and parameters from their full conditionals. EM instead takes the conditional expectation of the complete-data log-likelihood and maximizes it. The common simplification is the complete-data model; the inferential goals and the updates are different.